



Marble sorter

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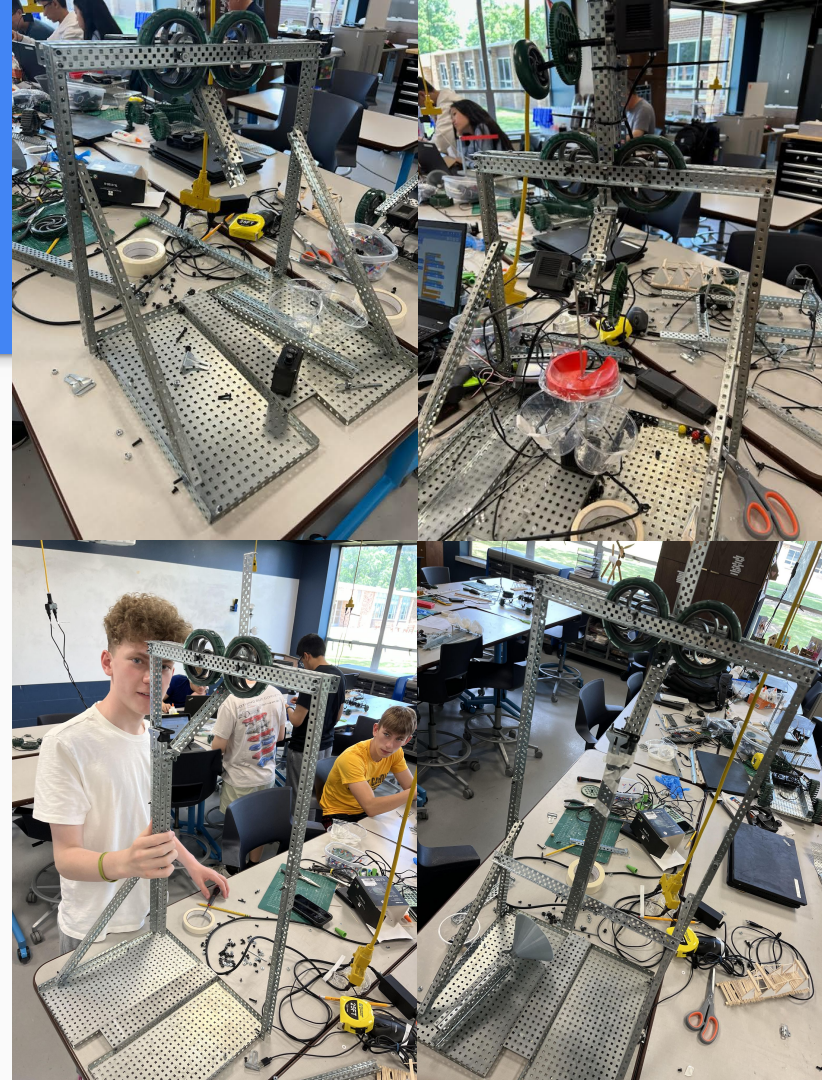
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Design process

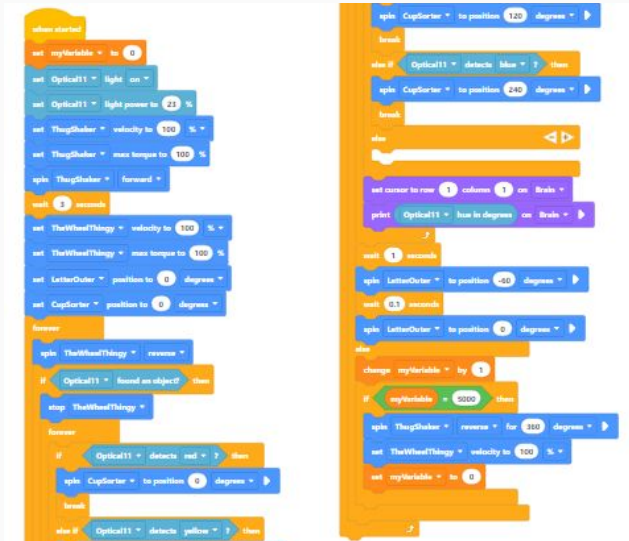
We were originally going to measure how high the marbles bounced to decide which cup they went in because all of our marbles bounce different heights. We later switched to sensing the color of the marbles and switched our marbles as well. We made lots of improvements to our machine and our code but more often than not changing the code made our machine not work as well as before.

Pictures

Pictures of us building the marble sorter



Code



Trials

Trial 1 - 13/15 (86.66%) | Trial 2 11/15 (73.33%) | Trial 3 14/15 (93.33%)

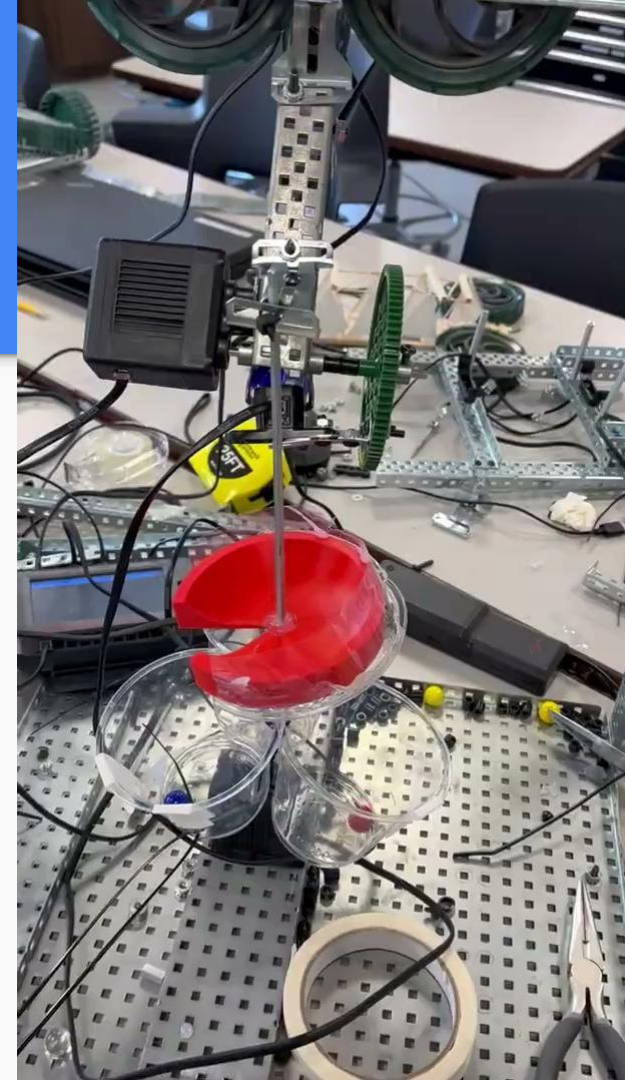
Trial 4 13/15 (86.66%) | Trial 5 13/15 (86.66%) | Trial 6 13/15 (86.66%)

Trial 7 15/15 (100%) | Trial 8 15/15 (100%) | Trial 9 14/15 (93.33%)

Trial 10 /15 (%)

Final process and success

We did surprisingly well. The only problems we were having was the sorter letting out more than one marble



Conclusion

We were successful two or three times out of ten. Our marble sorter was about 20-30% accurate. It is not too difficult to make a marble color sensor but it is difficult to make it accurate. Our design would've been perfectly accurate if the color of our marbles weren't so similar.

R/W/L

Right:

We successfully created a marble sorter that sorted 15 marbles into three groups.

Wrong:

We started with a different design. We were going to sort the marble by how much they bounced. We had to scrap this idea. This wasted a lot of time.

Learned:

We learned to think about and plan the design before you build it. This way you will save time and it will probably make the build better.